

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Total Marks: 25

Question Count: 5

Code of Conduct

I B Sandhya(192311292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sandhya

Assessment Tasks

Application Based Questions

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy.

How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application.
4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions.
5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide an example of a microservices-based application.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage

Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

ANSWERS

1.Cloud Auto-Scaling and Load Balancing for Seasonal Sales

Introduction

During seasonal sales such as **Black Friday**, **Diwali**, or **Amazon Great Indian Festival**, an e-commerce company experiences sudden spikes in customer traffic. Cloud technologies like **Auto-Scaling** and **Load Balancing** help maintain application performance, availability, and reliability during these high-demand periods.

How Auto-Scaling and Load Balancing Handle Traffic Efficiently

Auto-Scaling

- Automatically increases the number of application servers when traffic increases.
- Removes extra servers when demand decreases.
- Uses metrics such as CPU utilization, memory usage, or request count to make scaling decisions.
- Ensures sufficient computing resources are always available.

Load Balancing

- Distributes incoming user requests evenly across multiple servers.
- Prevents any single server from becoming overloaded.
- Detects unhealthy servers and redirects traffic to healthy ones.
- Improves application availability and response time.

How This Reduces Downtime and Improves User Experience

Reduces Downtime

- Additional servers are launched automatically during traffic spikes.
- Failed servers are removed from service without affecting users.
- Prevents server crashes caused by overload.
- Ensures continuous service availability.

Improves User Experience

- Faster page loading.
- Reduced checkout delays.
- Minimal buffering or timeout errors.
- Smooth shopping experience even during peak sales.
- Increased customer satisfaction.

Role of Monitoring Tools

Cloud monitoring tools continuously track the health and performance of the infrastructure.

They monitor:

- CPU usage
- Memory utilization
- Network traffic
- Response time
- Error rates
- Server health
- Number of active users

They provide:

- Real-time alerts for unusual activity.
- Automatic triggering of auto-scaling policies.
- Performance dashboards.
- Log analysis for troubleshooting.
- Early detection of failures.

Real-World E-Commerce Scenario

Consider **Amazon** during the **Great Indian Festival** sale.

- Millions of customers visit the website simultaneously.
- A **Load Balancer** distributes customer requests across many application servers.

- **Auto-Scaling** automatically launches additional servers as traffic increases.
- Monitoring tools detect high CPU usage and trigger scaling actions.
- If one server fails, traffic is redirected to healthy servers without interrupting customer shopping.
- After the sale ends, extra servers are automatically removed, reducing cloud costs.

This ensures:

- High availability
- Fast website performance
- Successful order processing
- Excellent customer experience

Advantages

- Handles sudden traffic spikes automatically.
- Reduces downtime.
- Improves application performance.
- Provides high availability.
- Optimizes cloud costs by scaling resources only when needed.
- Enhances customer satisfaction.

Conclusion

Cloud **Auto-Scaling** and **Load Balancing** enable organizations to efficiently manage unpredictable traffic spikes during seasonal sales. Auto-scaling dynamically adjusts computing resources based on demand, while load balancing distributes user requests evenly across available servers. Monitoring tools continuously track system health and automatically trigger scaling actions when needed. In real-world e-commerce platforms such as **Amazon**, this approach ensures **high availability, minimal downtime, fast response times, and a seamless shopping experience**, while also optimizing infrastructure costs.

2. Cloud Storage for Secure Backup and Disaster Recovery

Introduction

Organizations generate large volumes of backup data that must be stored securely and remain available during disasters. **Cloud storage solutions** provide high durability, data redundancy, security, and cost-effective storage compared to traditional on-premises storage systems.

How Cloud Storage Ensures Durability and Data Redundancy

Durability

- Cloud providers store multiple copies of backup data.
- Data is automatically checked for corruption and repaired if necessary.
- Backup data remains available even if hardware fails.

Data Redundancy

- Data is replicated across multiple disks, servers, or data centers.
- Many cloud providers also offer **cross-region replication** to protect against regional failures.
- If one storage location becomes unavailable, another copy is used automatically.

Benefits

- Prevents data loss.
 - Ensures high availability.
 - Improves reliability during hardware or regional failures.
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How Encryption and Access Control Protect Backup Data

Encryption

Data at Rest

- Backup files are encrypted using **AES-256** before or while being stored in the cloud.

Data in Transit

- Data is protected using **TLS/SSL** encryption while transferring between the organization and the cloud.

Access Control

- Use **Identity and Access Management (IAM)** to control user permissions.
- Implement **Role-Based Access Control (RBAC)** so users access only the data they need.
- Enable **Multi-Factor Authentication (MFA)** for added security.
- Maintain audit logs to track all backup access and modifications.

Benefits

- Protects sensitive information.
- Prevents unauthorized access.
- Ensures compliance with security regulations.

Cost Advantages Compared to Traditional Storage

1. **Pay-as-You-Go Pricing**
 - Pay only for the storage actually used.
2. **No Hardware Investment**
 - Eliminates the need to purchase storage servers and backup devices.
3. **Reduced Maintenance Costs**

- Cloud providers manage hardware, upgrades, and maintenance.
- 4. **Scalable Storage**
 - Increase or decrease storage capacity as needed without buying new equipment.
- 5. **Lower Disaster Recovery Costs**
 - Built-in backup and replication reduce the need for separate disaster recovery infrastructure.

Disaster Recovery Use Case

A **hospital** stores patient records and medical reports in cloud storage.

- Daily backups are automatically uploaded to the cloud.
- Data is encrypted and replicated across multiple regions.
- If the hospital's local data center is damaged by a flood or fire, the backup can be restored from the cloud.
- Medical staff can quickly recover patient records and continue providing healthcare services with minimal downtime.

Advantages

- High durability and reliability.
- Automatic data redundancy.
- Strong encryption and secure access control.
- Easy scalability.
- Cost-effective backup storage.
- Fast disaster recovery.
- Improved business continuity.

Conclusion

Cloud storage provides a secure and reliable solution for storing large volumes of backup data through **high durability**, **data redundancy**, and **cross-region replication**. Encryption, IAM, RBAC, and MFA protect sensitive backup information from unauthorized access. Compared to traditional storage systems, cloud storage reduces hardware, maintenance, and disaster recovery costs while offering flexible, pay-as-you-go scalability. In disaster recovery scenarios, organizations can quickly restore critical data from the cloud, ensuring **business continuity**, **minimal downtime**, and **data protection**.

3. CDN for Global Mobile Application Deployment

Introduction

A **mobile app company** deploying its application globally must ensure fast content delivery, low latency, and high availability **for users worldwide**. Content Delivery Networks (CDNs) **help achieve these goals by delivering content from servers located close to users**.

How CDNs Improve Performance and Reduce Latency

A **Content Delivery Network (CDN)** is a network of geographically distributed **edge servers** that cache and deliver application content.

CDN Working

- Frequently accessed content (images, videos, app files, APIs, etc.) is stored on edge servers.
- When a user requests content, it is served from the nearest edge server instead of the central server.
- This reduces the distance data must travel, improving speed.

Benefits

- Faster content delivery
- Lower network latency
- Reduced load on the origin server
- Improved application performance
- Better scalability during high traffic

How Edge Servers Enhance User Experience

Edge servers are located in multiple geographic regions.

They provide:

- Faster response times by serving nearby users.
- Reduced buffering for video streaming.
- Lower lag in online games.
- Faster app downloads and updates.
- High availability even during peak traffic.

Example

- A user in India receives content from a nearby CDN edge server in Asia.
- A user in Europe receives the same content from a European edge server.
- Both users experience fast loading times without accessing the main server.

How Cloud Providers Support Global Availability

Cloud providers offer worldwide infrastructure to ensure applications remain available.

Features include:

- Multiple global regions and availability zones.
- Content Delivery Networks (CDNs).
- Global Load Balancers.

- Auto Scaling.
- Cross-region data replication.
- Disaster recovery services.
- Health monitoring and automatic failover.

Benefits

- High availability
- Fault tolerance
- Low latency
- Better performance for global users
- Business continuity

Real-Time Streaming or Gaming Example

Consider **Netflix** or an online multiplayer game.

- Millions of users stream videos or play games simultaneously.
- Videos and game assets are cached on CDN edge servers worldwide.
- Users automatically connect to the nearest edge location.
- If one server or region fails, traffic is redirected to another healthy server.
- Auto-scaling adds additional servers during peak hours.

This results in:

- Smooth video playback
- Minimal buffering
- Low gaming latency
- Fast content loading
- Better user experience worldwide

Advantages

- Faster content delivery
- Reduced latency
- High availability
- Lower bandwidth usage
- Improved scalability
- Better reliability
- Enhanced user satisfaction

Conclusion

Content Delivery Networks (CDNs) play a vital role in global mobile application deployment by caching content on **edge servers** located close to users. This significantly reduces latency, improves performance, and decreases the load on origin servers. Cloud providers further enhance global availability through multiple regions, load balancing, auto-scaling, and disaster recovery services. In real-time streaming platforms like **Netflix** and

online gaming applications, CDNs ensure **fast content delivery, minimal buffering, low latency, and a seamless user experience** for users around the world.

4. Cloud Security, Compliance, and Data Sovereignty for a Financial Institution

Introduction

Financial institutions handle highly sensitive customer and transaction data. They must comply with regulations while ensuring data confidentiality, integrity, and availability. Cloud providers offer built-in security, compliance, auditing, encryption, and monitoring services to help organizations meet these requirements.

How Cloud Providers Support Regulatory Compliance and Auditing

Cloud providers support various international and industry standards, such as:

- **ISO 27001** (Information Security Management)
- **SOC 1 & SOC 2** (Security and Operational Controls)
- **PCI DSS** (Payment Card Industry Data Security Standard)
- **GDPR** (General Data Protection Regulation)

Compliance Features

- Regular security audits and compliance certifications.
- Automated compliance reports.
- Identity and Access Management (IAM).
- Role-Based Access Control (RBAC).
- Continuous security monitoring.
- Audit logs for all user and system activities.

Benefits

- Meets legal and regulatory requirements.
- Improves transparency.
- Simplifies compliance audits.
- Enhances trust and accountability.

Role of Logging, Monitoring, and Encryption

Logging

- Records all user logins, transactions, configuration changes, and access attempts.
- Maintains audit trails for investigations and compliance.

Monitoring

- Continuously tracks system performance and security events.
- Detects suspicious activities and unauthorized access.
- Generates real-time alerts for potential threats.

Encryption

Data at Rest

- Encrypts stored financial records using **AES-256** encryption.

Data in Transit

- Protects communication using **TLS/SSL** encryption.

Key Management

- Stores encryption keys securely using **Key Management Services (KMS)** or **Hardware Security Modules (HSMs)**.

Benefits

- Protects confidential financial information.
- Prevents unauthorized access.
- Ensures secure data transmission.
- Supports compliance requirements.

Ensuring Data Sovereignty

Data sovereignty means data must remain within a specific country or region to comply with local laws.

Organizations can ensure this by:

- Storing data only in approved cloud regions.
- Using region-specific storage and databases.
- Enabling data residency policies.
- Restricting cross-border data transfers.
- Applying access controls based on geographic location.
- Performing regular compliance audits.

Benefits

- Meets government regulations.
- Protects customer privacy.
- Reduces legal risks.

Example: Sensitive Financial Transactions

Consider an **online banking system**.

- Customers transfer money using a mobile banking application.
- Transaction details are encrypted before transmission.
- Data is stored in cloud databases located within the required country to satisfy data sovereignty regulations.
- Every login, transaction, and administrative action is logged for auditing.
- Monitoring systems detect unusual activities, such as multiple failed login attempts or suspicious transactions, and generate alerts.
- If suspicious behavior is detected, security teams can investigate using audit logs while customer data remains protected.

Advantages

- Strong regulatory compliance.
- Secure storage and transmission of financial data.
- Complete audit trails.
- Real-time threat detection.
- Protection against unauthorized access.
- Compliance with data sovereignty laws.
- Improved customer trust.

Conclusion

Cloud providers help financial institutions achieve **regulatory compliance** through certified security standards, auditing capabilities, and centralized identity management. **Logging, monitoring, and encryption** work together to protect sensitive financial information, detect security threats, and maintain detailed audit records. By storing data in approved geographic regions and enforcing data residency policies, organizations can satisfy **data sovereignty** requirements. These measures ensure secure financial transactions, regulatory compliance, business continuity, and enhanced customer confidence.

5.Containerization for Application Deployment

Introduction

Containerization allows developers to package an application along with its dependencies into a lightweight **container**. Containers provide consistent execution across development, testing, and production environments. Container orchestration platforms such as **Kubernetes** simplify deployment, scaling, and management of containerized applications.

Containers vs Virtual Machines

Feature	Containers	Virtual Machines (VMs)
Operating System	Share the host operating system	Each VM has its own operating system
Startup Time	Starts in seconds	Takes minutes to boot

Feature	Containers	Virtual Machines (VMs)
Performance	Faster and lightweight	Slower due to full OS overhead
Resource Usage	Uses less CPU, memory, and storage	Uses more CPU, memory, and storage
Portability	Easily portable across platforms	Less portable due to larger images
Isolation	Process-level isolation	Full hardware-level isolation

Advantages of Containers

- Lightweight
- Faster startup
- Better resource utilization
- Easier deployment
- Consistent application environment

How Kubernetes Simplifies Deployment and Scaling

Kubernetes is a container orchestration platform that automates the deployment, management, and scaling of containers.

Deployment

- Automatically deploys containerized applications.
- Restarts failed containers.
- Performs rolling updates without downtime.
- Maintains the desired number of application instances.

Scaling

- Automatically increases or decreases the number of containers based on workload.
- Uses **Horizontal Pod Autoscaler (HPA)** to scale applications.
- Balances traffic among available containers using built-in load balancing.

Benefits

- High availability
- Automatic recovery
- Efficient resource utilization
- Simplified application management
- Reduced manual intervention

Portability Across Different Cloud Environments

Containers package the application and all required dependencies into a single unit.

This enables applications to run consistently across:

- Private Cloud
- Public Cloud
- Hybrid Cloud
- Multi-Cloud environments

Benefits

- No dependency conflicts.
- Easy migration between cloud providers.
- Reduced vendor lock-in.
- Consistent behavior across environments.
- Faster development and deployment.

Example of a Microservices-Based Application

An **Online Shopping (E-Commerce) Application** can be built using multiple microservices.

Microservices

- **User Service** – Manages user accounts and authentication.
- **Product Service** – Handles product catalog and inventory.
- **Order Service** – Processes customer orders.
- **Payment Service** – Handles online payments.
- **Notification Service** – Sends email and SMS notifications.

Each microservice runs in its own container and can be deployed, updated, and scaled independently using Kubernetes.

Example:

- During a sale, the **Product Service** and **Order Service** receive heavy traffic.
- Kubernetes automatically scales only these services without affecting the others.
- If one container fails, Kubernetes restarts it automatically while the remaining services continue running.

Advantages

- Faster deployment.
- Better scalability.
- Efficient resource utilization.
- High availability.
- Easy maintenance and updates.
- Cloud portability.
- Supports Continuous Integration and Continuous Deployment (CI/CD).

Conclusion

Containers provide a lightweight and efficient alternative to virtual machines by sharing the host operating system, resulting in **better performance** and **lower resource usage**.

Kubernetes simplifies container management through **automated deployment**, **load balancing**, **self-healing**, and **auto-scaling**. Containerized applications are highly portable and can run consistently across different cloud providers, reducing vendor lock-in. In a **microservices-based e-commerce application**, containers and Kubernetes enable independent deployment, scaling, and maintenance of services, ensuring **high availability**, **flexibility**, and **efficient application management**.